“Damn that DAWN!” you think to yourself. “She always answers my questions, but she never volunteers crucial information. She never warns me when I’m going wrong!” You know the reason, of course. Human life would lose its zest and richness if computers were permitted to give warning whenever a mistake was being made. Back in 2032 the World Council passed a law that a Hobson block preventing such warnings must be embedded in all computers. As much as she might wish, DAWN cannot bypass her Hobson block.

Suppressing your exasperation, you rotate your capsule and begin a careful sequence of forward blast, inward spiral, coast, forward blast, inward spiral, coast, forward blast, inward spiral, coast, which takes you from 3 horizon circumferences to 2.5 to 2.0 to 1.6 to 1.55 to 1.51 to 1.505 to 1.501 to ... What frustration! The more times you blast and the faster your resulting coasting, circular motion, the smaller becomes your orbit; but as your coasting speed approaches the speed of light, your orbit only approaches 1.5 horizon circumferences. Since you can’t move faster than light, there is no hope of getting close to the horizon itself by this method.

Again you appeal to DAWN for help, and again she soothes you and explains: Inside 1.5 horizon circumferences there are no circular orbits at all. Gravity’s pull there is so strong that it cannot be counteracted by any centrifugal forces, not even if one coasts around and around the hole at the speed of light. If you want to go closer, DAWN says, you must abandon your circular, coasting orbit and instead descend directly toward the horizon, with your rockets blasting downward to keep you from falling catastrophically. The force of your rockets will support you against the hole’s gravity as you slowly descend and then hover just above the horizon, like an astronaut hovering on blasting rockets just above the Moon’s surface.

Having learned some caution by now, you ask DAWN for advice about the consequences of such a strong, steady rocket blast. You explain that you wish to hover at a location, 1.0001 horizon circumferences, where most of the effects of the horizon can be experienced, but from which you can escape. If you support your capsule there by a steady rocket blast, how much acceleration force will you feel? “One hundred and fifty million Earth gravities,” DAWN replies gently.

Deeply discouraged, you blast and spiral your way back up into the belly of the starship.